

4(a)	much / AW, rough endoplasmic reticulum / rough ER / RER for, polypeptide / protein / collagen, synthesis ; A for translation many mitochondria provide, energy / ATP, for, polypeptide / protein / collagen, synthesis ; large nucleus indicates, active cell / (much) transcription ;	2
4(b)(i)	<i>any two from</i> in trachea ; in, bronchus / bronchi ; <i>if only one correct structure named, allow one qualification mark</i> <i>trachea</i> C-shaped / incomplete, rings ; surround smooth muscle ; <i>bronchus</i> plates / irregular ;	2

Question	Answer	Marks
4(b)(ii)	<i>any two from</i> keep airways open ; provides support ; allow flexibility ; <i>allow described e.g. bending neck, swallowing food</i> rings allow, lengthening / widening, during, breathing in / inspiration / inhalation ;	2
4(c)(i)	peptide (bond) ;	1
4(c)(ii)	<i>any three from</i> collagen, structural / fibrous, protein <i>or</i> collagen gives strength / flexibility ; glycine / gly, small / smallest, amino acid ; A has H as, R-group / side chain glycine / gly, regular / every third amino acid ; (so) triple helix tightly packed / three polypeptides closely associated / AW ; <i>ref. to</i> (peptide bond) NH of gly can form hydrogen bond with (peptide bond) C=O of adjacent amino acid (in other polypeptide) ;	3

Question	Answer	Marks
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